

as a unique identifier known as the hash of the previous block. Collectively, these blocks create a permanent series of records classified as a chain of data in a secure and consistent manner.

Key features and characteristics of blockchain include:

Decentralisation: No government body or individual organisation controls the data.

Transparency: All parties in a transaction can see and confirm the transaction.

Immutable: Once data has been created and recorded, it is impossible to alter it.

Secured: Data is secured against being altered through a series of cryptographic processes.

Consensus mechanism: The process of authenticating and validating transactions requires approval from all members of the common network.

For example, blockchains are used by banks to provide secure, unchangeable records of transactions. Each record added to the blockchain is permanent, allowing parties to trust each other, thereby reducing the risk of fraud.

The system architecture of the blockchain consists of many blocks lined up chronologically, containing both transaction records and the digital signature of each member in the network. This forms an unbroken chain of linked blocks, where each block contains transaction and membership information pertaining to multiple users within the network. Hence, the blockchain has a strong inherent immunity to manipulation of its data and can be trusted by all parties involved.

Architecture: Blocks and cryptographic hashing

A blockchain consists of three major elements: transaction data, date/time stamp and a cryptographically created hash. The hash acts as a unique digital representation of the preceding blocks and is created by applying cryptographic algorithms to the information within each block. By linking each block to another via hashes, it is possible to determine whether any individual piece of information in an individual block has changed. This breaks the connection between any given block and all its successors, making it obvious that someone has tampered with a given block.

The mechanism of chain formation

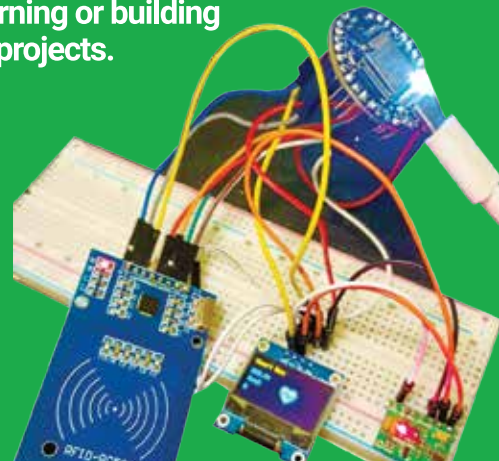
Each new block added in the blockchain builds up the chain incrementally and in succession. Transactions go through verification via nodes on the network before they are bundled together in a single block once validated by the network. That block is then incorporated into the previously existing chain of blocks, thus creating an everlasting chronological record of all transactions ever made and continually making that updated and verified information accessible to all parties involved.

As an example, in a supply chain system, steps like manufacturing, delivery, shipping, and delivery can be

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